Improved Ingredients-based Recipe Recommendation Software using Machine Learning

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***Abstract* — in this paper we propose an improved recipe recommendation system that employs image recognition of food ingredients. The system is currently a mobile application that performs image recognition on uploaded or camera-captured images and recommends recipes containing the recognized ingredients. We used the ResNet-V2 architecture to build a convolutional neural network model for image recognition, which was able to identify 33 different food ingredients with an accuracy rate of 89%. The recommendation system uses the identified ingredient labels, as well as user preferences and restrictions, to display a list of recipes containing the identified ingredients. This feature allows users to discover new and exciting recipes based on the ingredients they currently have at home, without having to worry about dietary restrictions or other preferences. Overall, our system provides a convenient and personalized way for users to discover and prepare delicious meals based on their unique needs and preferences.**

Keywords—Artificial Intelligence, Machine Learning, ResNet neural network, Recipe recommendation, Preference, Content-based

I. Introduction

Recommender systems that leverage user interests and preferences have become ubiquitous in various websites and applications, such as Netflix, and YouTube [1]. However, cooking websites and applications that allow users to search for recipes based on keywords or ingredients may not account for user-specific constraints, such as dietary restrictions or food allergies [2].

To address this issue, we propose a recipe recommendation system that utilizes image recognition technology to identify available ingredients and suggest recipes accordingly. The main objective of our proposed system is to provide users with personalized and relevant recipe recommendations based on their available ingredients, dietary preferences, and cultural or regional eating habits. Advanced machine learning techniques like Conventional neural network and ResNet are employed to learn from user eating habits and preferences, as well as the ingredients they are looking for. Users can search for recipes using keywords or by utilizing the system's image recognition technology.

Cooking is an essential part of daily life, but many people lack experience in preparing meals or struggle to find recipes that meet their needs. Traditional recipe search methods based on keywords can be time-consuming and frustrating, particularly for those with dietary restrictions or limited access to certain ingredients due to cultural or regional eating habits. Furthermore, users may not be able to find recipes that fit their needs or preferences, resulting in disengagement and dissatisfaction [3].

The proposed system aims to address these issues by providing a more personalized and efficient recipe search experience that saves time and reduces frustration. By leveraging image recognition technology and machine learning algorithms, and to improve the accuracy of the software and prevent overfitting techniques like data augmentation, batch normalization are used and our system can suggest personalized recipe recommendations that promote healthy and diverse eating habits among users.

The motivation for this research is to improve user engagement and satisfaction while also addressing the challenges associated with finding and preparing meals.

The Next section of this paper covers related work and existing approaches. Section III presents our methodology, including an overview of the system, processing steps and architecture. Additionally, we describe ingredients classification approaches and improvements to the

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recommendation system. Section IV provides an experimental analysis, evaluating the classification models' performance and comparing results. In section V, we conclude the paper by summarizing key findings and suggesting future research directions.

II. Related Work

Recent years have seen a growing interest in recipe recommendation, ingredient detection, and food detection, which are all based on deep learning techniques for object detection and classification. There are several deep learning models and methods available for these tasks, including Support Vector Machine (SVM) classifier, and K-nearest neighbors. These methods are widely recognized and established in the field [4].

M. Ueda et al [5] have proposed a text-based method for cooking recipe recommendation based on the user's preference represented as a set of keywords. Their method uses a probabilistic model trained on a dataset of recipe data and user preferences, where the recipe data is represented as a bag-of-words and the user preference is represented as a vector of binary values. To recommend a recipe to the user, the method computes the probability of each recipe given the user's preference vector and returns the recipe with the highest probability.

Keiji Yanai et al, [6] have proposed a real-time cooking recipe recommendation system using two types of real-time image recognition methods, the recommendation is based on the conditions of ingredients on amounts, nutrition and prices which is recognized from a short video which runs on a consumer smartphone using SVM technique with accuracy 83.93%.

Dubey and Jalal [7] have presented a method for the classification of fruits and vegetables, which involves detecting and segmenting objects, extracting relevant features, and training SVM models. Once the classification is done, the system can suggest recipes based on the recognized fruits and vegetables.

Mona Mishra, and Yifan Gong [8] have proposed a recipe recommendation system employing image recognition of food ingredients, specifically vegetables using Convolutional neural network CNN and SVM models for a website with accuracy 62.9%.

Our system is motivated by the approaches used in these papers to make a recommendation system based on ingredients the user currently has, also consider his/her preference and if there are any diet or food restrictions, also using the deep learning methods to classify different categories of ingredients. In the following section, we will discuss our proposed model.

III. Methodology

A. Overview

The objective of this system is to help users who suffer from not being able to decide what recipes they should prepare on daily basis by making it easier for them by providing a mobile application that requires only opening the camera and taking a picture of the ingredients available at hand and then proposing multiple recommendations from a saved cooking recipes dataset that contains thousands of recipes instantly while putting in consideration the user's preferences and history.

The system accomplishes this by identifying food ingredients in photos captured by its built-in cameras and then searching its cooking recipe databases for recipes that include those recognized food ingredients. In addition, the system also takes into account the user's dietary restrictions by referring to its database of preferences.

For the ingredients recognition, transfer learning is used by applying a pre-trained ResNet50 [9] model to a database with 33 different classes of food and top ranked, content based on preferences and collaborative filtering by ranking techniques are used to do the prediction.

B. Processing steps

In this subsection, we describe the flow of how to use the proposed system from taking photos of food ingredients to watching the recipe pages a list of recipes.

1. When registering for an account on the application for the first time you are asked to fill in if you have any preferences in your meals like spicy, vegan.
2. Once logged in, users can either search by camera or by text. If he chooses to search by camera, he is given

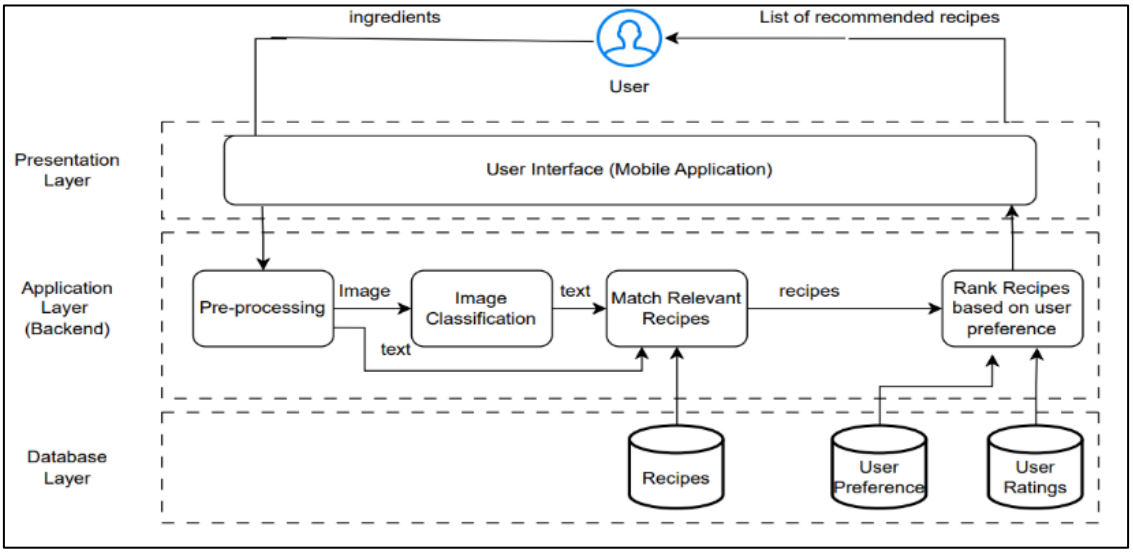

Fig. 1. System architecture

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the option to either take a real-time picture or upload one from the gallery.

3. The system then shows the identified ingredients to assure they were correctly identified and then takes him to a page with a list of recipes and all their information like steps, ingredients, cooking time that includes the specified ingredients and taking in consideration his preferences.
4. The user can also choose to search by text by entering ingredients names and getting the same results.

C. System Architecture

The proposed system comprises three layers as shown in Fig.1:

Presentation Layer (Frontend): The mobile app component serves as the presentation layer of the system, responsible for handling user interactions and displaying information to the user which is the list of recommended recipes according to ingredients, preferences and ratings.

Application layer (Backend): the application layer of the system encompasses various components such as authentication, preprocessing of images and ingredients, classification, and recommendation of recipes. These components are responsible for executing the system's business logic, handling user requests, and generating customized recipe suggestions by considering factors such as preferences, search history, recommendations, ratings, and favorites.

Data Layer: The Firebase database platform serves as the data layer of the system, responsible for storing and retrieving user data, including user preferences, ratings, and favorite recipes, as well as the recipes dataset.

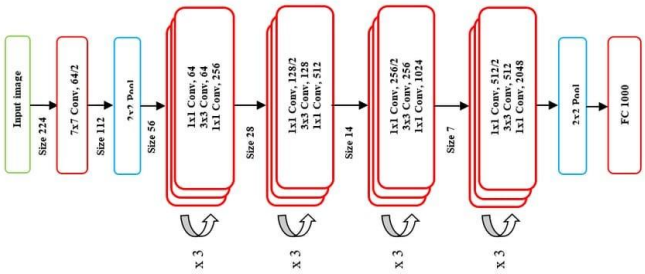

Fig. 2. ResNet50 architecture [7].

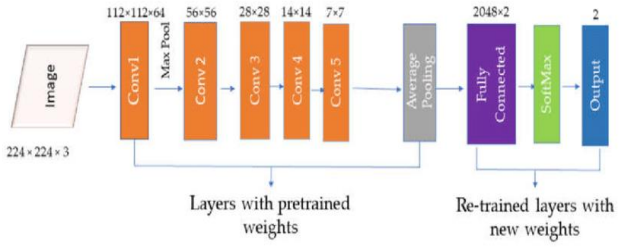

Fig. 3. Improved ResNet architecture.

IV. Ingrednits Classfication Approches

In this section, we provide a method for identifying different categories of food ingredients. This method takes into account image features, image attributes, image representation, and image classifier.

A. Overall architecture

Our recipe recommendation system utilizes a ResNet50V2 convolutional neural network (CNN) [10] model for food ingredient classification. We chose this model due to its excellent performance in image classification tasks and its ability to handle input images of different sizes. We replaced the original ImageNet classifier at the top of the ResNet50V2 model with our own 33-class SoftMax layer to classify food ingredients.

The ResNet50V2 model in Fig. 2 is pre-trained on ImageNet and includes 50 convolutional layers with residual connections, which help to mitigate the vanishing gradient problem. To adapt the model for our task, we excluded the original classifier and added our own global average pooling layer to reduce the spatial dimensions of the output. We also applied batch normalization to normalize the activations of the previous layer and reduce overfitting, followed by a dropout layer to further regularize the model.

The final architecture of our model is shown in Fig. 3 which consists of an input layer that takes images of size 224x224x3, followed by the ResNet50V2 base model, the global average pooling layer, batch normalization, and dropout layers, and two fully connected layers with ReLU activation and dropout. The output layer is a dense layer with 33 units and a SoftMax activation function to predict the probability of each food ingredient class.

B. Preprocessing

Convolutional neural networks [11] require inputs of the same size, so all food ingredient images need to be resized to a consistent size before being inputted to the CNN. For the ResNet50 model, we targeted an input size of 224 x 224 pixels. However, our dataset included images of varying sizes and shapes that were collected from different sources, including free dataset from GitHub and images captured with our own camera to ensure there's a variety in the data images. Training CNN on unprocessed images can lead to poor classification performance.

Therefore, we resized the images to 224 x 224 pixels and removed any unnecessary objects from the images to ensure that the food ingredients were the main focus of the images. This preprocessing step helped to improve the classification accuracy of our model.

C. Data Augmentation

To preprocess the input images, we applied data augmentation techniques using the data augmentation function from the Keras preprocessing module. This function randomly applies transformations to the input images, such as rotation, zooming, and flipping, to increase the diversity of the training data and prevent overfitting.

D. Optimizer and learning rate

The choice of optimizer, learning rate, activation function, and loss function are critical for training deep neural networks. In our recipe recommendation system, we use the Adam optimizer with a learning rate of 0.001 and learning rate decay. The Adam optimizer is an adaptive learning rate optimization algorithm that is well-suited for deep learning applications. It uses gradient descent with momentum and root mean square propagation (RMSprop) to update the model weights. The learning rate determines the step size at which the model weights are updated during training. We chose a

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learning rate of 0.001 based on experimentation and found that it worked well for our recipe recommendation system.

The output layer of our system is a dense layer with 33 units and a SoftMax activation function, which converts the output to a probability distribution over the 33 food ingredient classes. The SoftMax function ensures that the output probabilities sum up to 1, which is necessary for interpreting the output as a probability distribution. The class with the highest probability is selected as the predicted class.

To train our model, we use the categorical cross-entropy loss function, which is commonly used in multi-class classification tasks. The categorical cross-entropy loss measures the dissimilarity between the predicted probability distribution and the true probability distribution of the classes. It is defined as:

where N is the number of samples, C is the number of classes, y is the true probability distribution (one-hot encoded) and $\hat{y}$ is the predicted probability distribution.

Overall, the Adam optimizer with a learning rate of 0.001 and learning rate decay, along with the SoftMax activation function and categorical cross-entropy loss function, provide a good combination for training our recipe recommendation system. However, these parameters can be further tuned based on the task requirements and the data distribution.

V. Improved Recommendation System

As mentioned, our aim is to recommend recipes to users from recognized ingredients taking into consideration the preference, so the recommendation engine is a combination of content based and collaborative based filtering to provide users accurate recommendations.

A. Weighted Average Score Technique

Weighted Average Score technique [12] was used to find the top-rated recipes in the dataset by previous users. It is represented in the following function.

$$w = \frac{Rv+Cm}{v+m} \tag{1}$$

Where R is the average rating of the recipe, v is the number of ratings for the recipe, C is the mean of aggregate ratings of all the recipes, m is the minimum ratings required to be listed in the chart of recipes that are considered.

B. Content-based filtering

Content based filtering [13] is the recommendation technique that suggests items based on the features of the items itself and it provides personalized recommendations to users.

To view the recipe recommendations for given ingredients by simply typing them or recognizing them, the recommendation engine was first made by a content-based algorithm which analyzes the recipe content, including ingredients, cooking method, and cuisine type, to suggest similar recipes that match a user's preferences and restrictions.

To represent the ingredients in a meaningful way, we used the Term Frequency Inverse Document Frequency (TF-IDF) [14] algorithm, which assigns weight to each ingredient in a recipe based on its frequency in the recipe and rarity in the overall dataset.it is represented in the following function.

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right) \tag{2}$$

Where $tf_{i,j}$ is the number of occurrences of i in j, df_i is the number of recipes containing i and N is the total number of recipes.

We also employed cosine similarity [15] to measure the similarity between two vectors, which is well-suited for sparse data common in recommendation systems. The cosine similarity score ranges from -1 to 1, where a score of 1 indicates two recipes are identical in terms of their ingredients, and a score of -1 indicates no common ingredients.

$$\cos(\theta) = \frac{A.B}{||A||.||B||} = \frac{\sum_i^n A_i B_i}{\sqrt{\sum_i^n A_i^2}\sqrt{\sum_i^n B_i^2}} \tag{3}$$

The recipe recommendations are sorted by ranking them based on two factors: the number of ingredients that match the user's search query and the user's preference. Recipes that contain more of the searched ingredients will be given higher priority, while recipes that align with the user's preferences will also receive a boost in ranking.

C. Collaborative-based filtering

Collaborative based filtering [16] is the recommendation technique that uses similarities between users and items to provide recommendations for users with similar preferences.

Collaborative filtering approach is used for building recommendation systems relying on ratings and behavior of other users in the system to suggest most relevant items to the user.

To address the limitations of content-based filtering, we further incorporated user-based collaborative filtering that considers the user's preferences, favorites, and ratings, as well as similarities between users and items. Collaborative filtering can be used to identify users who have similar tastes in food and recommend recipes that have been liked or rated highly by these similar users.

This approach allows us to consider not only the items themselves, but also the behavior and preferences of users, resulting in more accurate recommendations that satisfy the users.

By leveraging the preferences of multiple users, collaborative filtering can provide more accurate and personalized recommendations, even for users with limited or incomplete data.

VI. Experimental analysis

A. Dataset

As for the recipes dataset , we used the Food.com dataset from Kaggle [17], that contains thousands of recipes, which offers various details on each recipe including cooking hours, servings, ingredients, nutrition, instructions, and more. Additionally, a variety of cuisines and meal kinds are covered. This helped us to identify user preference.

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To classify the ingredients provided by the user, we got a customized food ingredients dataset from GitHub that contains a collection of food ingredient images. The dataset contains 3,500 images of 33 different various food ingredients categories, such as fruits, vegetables, meats, fish, and chicken. We used this dataset as input data for our classification model to recommend recipes based on the user's input.

The preference dataset used in the system comprises the preferences selected by users during the application's signup process that results in the variety of different recipes for each user. The dataset includes 10 distinct types of preferences, namely: spicy, lactose-free, low protein, high protein, healthy, vegan, very low carbs, high carbs, low cholesterol, and high/low blood pressure.

B. *Classfication Model Performance*

As we mentioned, we have a dataset of a total of 3500 images. So, we took 80% of the images for the training, 10% for testing, and the rest of the 10% for validation. After running 50 epochs using transfer learning and applying early stopping, our proposed CNN-based ResNet50 model achieved a 95.69% accuracy for the training dataset. And 89.94% accuracy on the validation dataset. After analyzing the results and confusion matrix, it can be stated that the CNN model has acceptable performance for classifying food ingredients. The CNN model's performance is shown below in Fig 4.

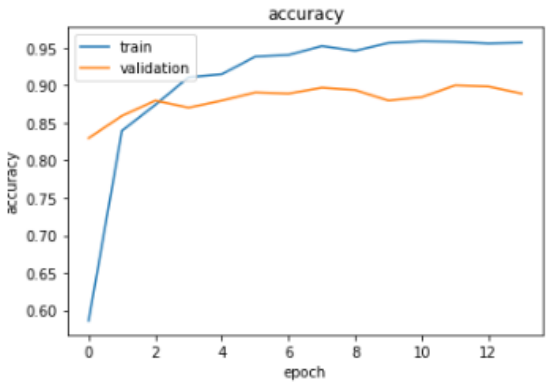

Fig. 4. Training and Validation accuracy

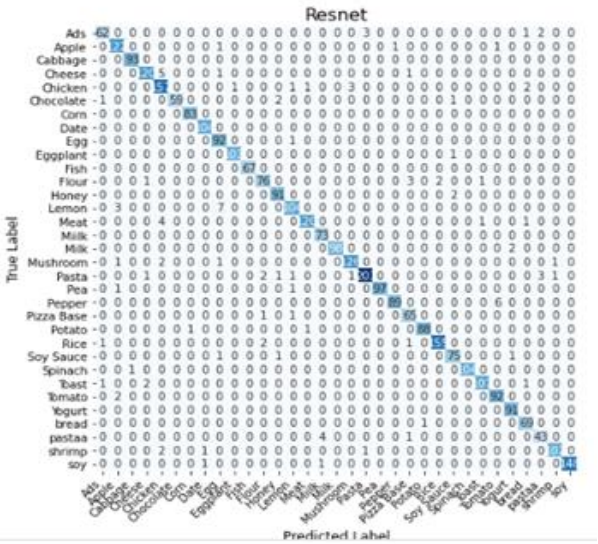

Fig. 5. Confusion metrics of ResNet Model

C. *Classfication Results Comparison*

We compared three different CNN models - VGG16 , MobileNet50 , and ResNet50 - to select the best-performing model [18].

Our results showed that ResNet50 outperformed the other models, achieving an accuracy of 90% on the test set. VGG16 achieved an accuracy of 60%, while MobileNet50 achieved an accuracy of 85%. The accuracy of the CNN model was 55%.

To further improve the performance of our model, we applied data augmentation techniques, including image rotation, shifting, and flipping. We also used techniques like dropout, early stopping, and batch normalization to avoid overfitting.

We visualized the performance of our model using confusion matrices and precision-recall curves illustrated in Fig. 5, which showed that our model performed well in identifying most of the ingredients.

The results of our study demonstrate that using a CNN architecture with transfer learning can effectively classify food ingredients. ResNet50 outperformed the other models due to its deeper architecture and better ability to capture high-level features in images.

Overall, our study shows the potential of using deep learning techniques for food ingredient classification, which can be used in recipe recommendation systems to provide personalized and relevant recipe suggestions to users based on their ingredient preferences. Future work could involve exploring other CNN architectures or combining multiple models to improve the accuracy of the classification task.

D. *Recommender Evaluation*

Mean Average Precision (MAP) and Mean Average Recall (MAR) are widely used evaluation metrics for assessing the performance of a recommender system. These metrics provide a more comprehensive evaluation of the system's performance beyond just accuracy or precision measures [19].

MAP measures the average precision of the top-K recommended items over all users. It takes the average of the precision values for each user, where precision is the proportion of relevant items among the top-K recommended items.

MAR measures the average recall of the top-K recommended items over all users. It takes the average of the recall values for each user, where recall is the proportion of relevant items that were recommended among all relevant items for the user.

Using our recommender, if the user inputs ingredients such as shrimp and pasta, the resulting recommendation list will be sorted in descending order from the most relevant to the least relevant recipes.

The sorting is based on a score that is calculated for each recipe, where the score represents the number of ingredients in the recipe that match the user's input. The higher the score, the more relevant the recipe is to the user's input. By sorting the recommendation list in this way, the most relevant recipes

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will be displayed at the top of the list, making it easier for the user to find recipes that match their preferences.

VII. Conclusion

In this paper, we have developed an improved recipe recommendation system that utilizes deep learning techniques to classify food ingredients and provide personalized recipe suggestions to users. Our study showed that using a convolutional neural network (CNN) with transfer learning can classify food ingredients with high accuracy. We compared three different CNN models; VGG16, MobileNet50, and ResNet50. We have found that ResNet50 outperforms the other models and can achieve an accuracy of 89% using the same test set.

We also implemented data augmentation techniques, dropout, early stopping, and batch normalization to improve the performance of our model and avoid overfitting. Our study shows the potential of using deep learning techniques for food ingredient classification and recipe recommendation, which can provide personalized and relevant recipe suggestions to users based on their ingredient preferences.

Further studies can be carried out to improve the model's performance by exploring other CNN architectures or combining multiple models to improve the accuracy of the classification task and add more features to the software to enhance user experience and provide more value to users.

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